

Z-TEST USING PYTHON

AIM

To conduct a Z test in Python to check whether the average weight of a species of birds differs from 150 grams.

PROGRAM

```
import numpy as np
import scipy.stats as stats

# Define the sample data (hypothetical weights in grams)
sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,
                        151, 150, 149, 152, 151, 148, 150, 152, 149, 150,
                        148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis
population_mean = 150

# Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1) # sample standard deviation

# Number of observations
n = len(sample_data)

# Calculate the Z-statistic
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

# Calculate the p-value (two-tailed test)
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

# Print results
print(f"Sample Size: {n}")
print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level
alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly "
          "different from 150 grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant "
          "difference in average weight from 150 grams.")
```

OUTPUT

```
Sample Size: 30
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from
150 grams.
```

RESULT

Thus the Z test was performed using Python. The p-value (0.5218) is greater than 0.05, so the null hypothesis is not rejected: there is no significant difference between the average weight of the birds and 150 grams.